

Trainings ▾

Check for Correct Component

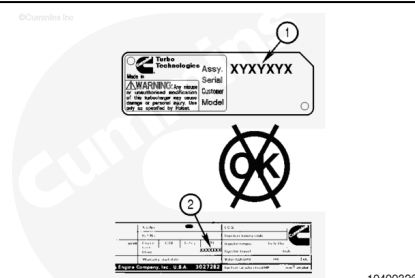
All Applications Except Marine and Rail

Check the assembly number (1) on the turbocharger dataplate with the turbocharger specified in the engine Control Parts List (CPL).

The CPL number for each engine is listed on the engine dataplate (2).

Note :

If an incorrect turbocharger was installed, remove it.



Preparatory Steps

All Applications Except Marine and Rail

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : This procedure applies to a turbocharger with a water-cooled bearing housing.

- Remove the intake and exhaust pipes from the turbocharger. Refer to the OEM service manual.
- Remove the air crossover. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

- Remove the oil drain hose. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-046.html)
- Remove the oil supply hose. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Remove the coolant supply hoses. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html)

Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)
- Remove the air crossover pipe. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)

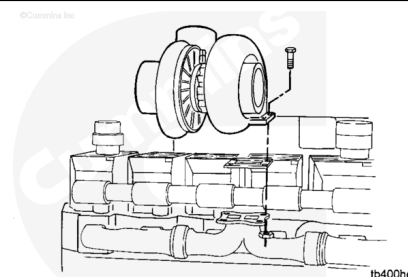
- Remove the exhaust piping. See equipment manufacturer service information.
- Remove the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-046.html>)
- Remove the turbocharger oil return tube. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-045.html>)

Remove

All Applications Except Marine and Rail

Remove the mounting capscrews, nuts, and the turbocharger.

Remove and discard the gasket.

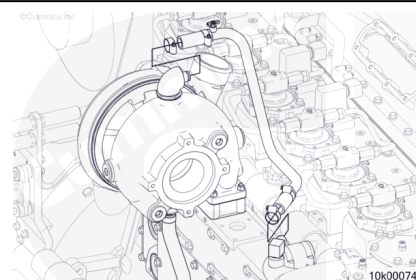


Marine Applications

Loosen clamps from turbocharger coolant return line then slide clamps to side.

Remove P-clip.

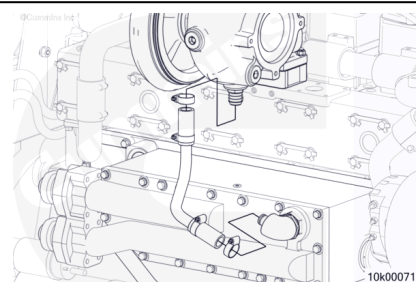
Remove turbocharger coolant return line from engine.



For engines with auxiliary cooler:

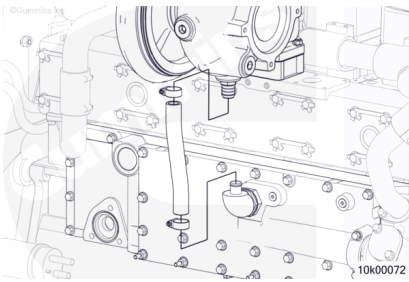
Loosen clamps from turbocharger coolant supply line then slide clamps to side.

Remove turbocharger coolant supply line from engine.



For engines without auxiliary cooler:

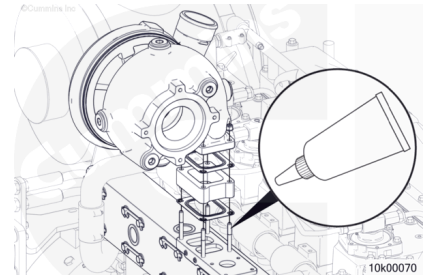
Loosen clamps from turbocharger coolant supply hose and remove hose.



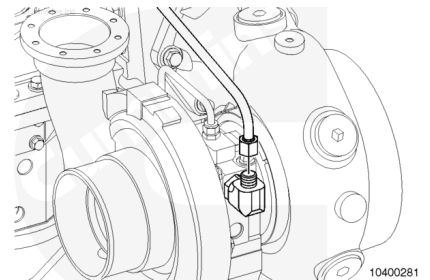
Remove the four nuts and washers from the turbocharger to exhaust manifold flange.

If equipped, remove turbocharger spacer.

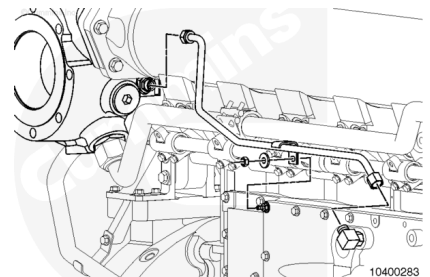
Remove turbocharger and discard gasket(s).



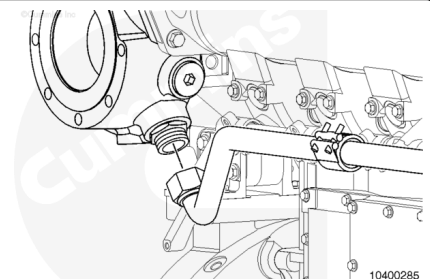
Remove the turbocharger bearing housing coolant supply line.



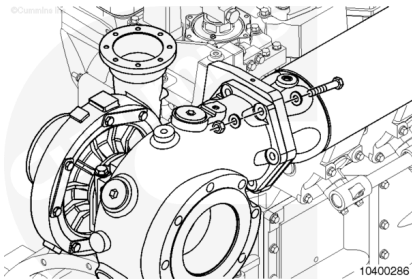
Remove the turbocharger bearing housing coolant return line.



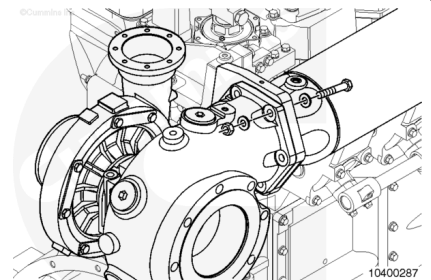
Remove the turbocharger turbine housing coolant return line from the turbocharger and coolant bypass line.



Remove the six capscrews, washers, and nuts from the turbocharger to exhaust manifold flange.

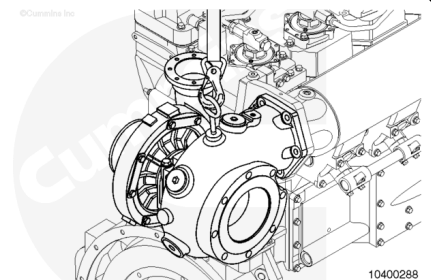


Remove the three capscrews from the support bracket to compressor housing.



Install a lifting eye bolt at the turbocharger turbine housing and attach an approved lifting device.

Support the weight of the turbocharger.

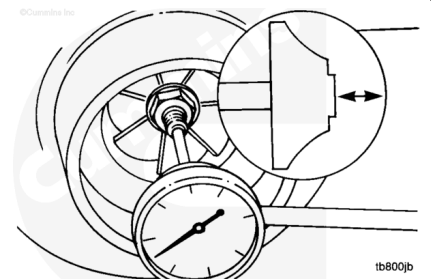


Inspect for Reuse

Check the exterior of the turbocharger for damage.

Check the wheel for fretting and broken vanes.

Use dial depth gauge, Part Number ST-537, or a dial indicator to measure the axial motion (end-to-end).



Axial Endplay

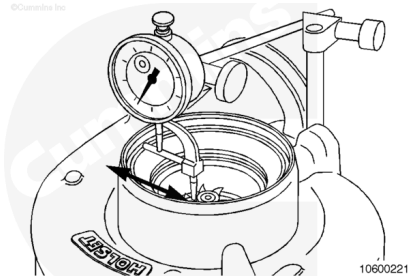
	mm		in
Holset® HX82 Turbocharg er Compress or Inlet	0.025	MIN	0.001
	0.152	MAX	0.006

Make sure movement is within MIN/MAX Total Indicator Reading (TIR) values shown above.

Note : If the end clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. See the Turbo Repair Manual HC5A/HX80/82/85/86, Bulletin 3580999, for rebuild instructions.

Measure the radial clearance (side-to-side) at the compressor impeller nose using a dial gauge.

Radial Clearance			
	mm		in
Holset® HX82 Compress or Impeller	0.254	MIN	0.01
	0.787	MAX	0.031



Make sure movement is within MIN/MAX Total Indicator Reading (TIR) values shown above.

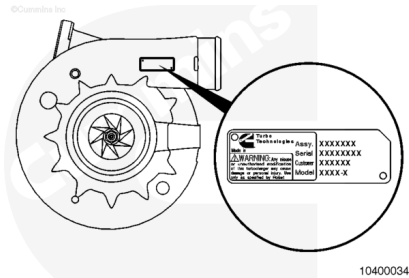
Note : If the radial clearance exceeds the specifications, the turbocharger must be replaced or rebuilt. Contact a Cummins® Authorized Repair Location. See the Turbo Repair Manual HC5A/HX80/82/85/86, Bulletin 3580999, for rebuild instructions.

Install

All Applications Except Marine and Rail

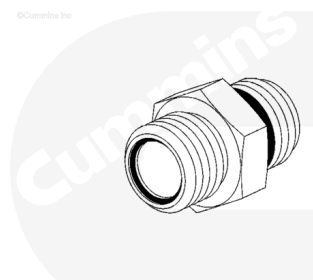
Cummins Turbo Technologies™ turbocharger(s) are used on the QSK19 engine. See the data tag on the turbocharger to determine the model.

The name is also cast on the housing of a turbocharger.



All turbocharger fittings are the flat face o-ring type fittings. Be sure the o-ring is in place before attaching the hose(s) to the fittings.

The Cummins Turbo Technologies™ turbocharger uses a 9/16-18 UNF, straight thread, o-ring type oil supply fitting.



Apply an anti-seize compound to the mounting capscrews.

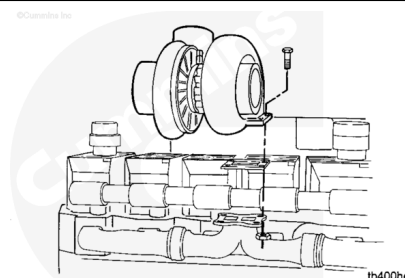
Install the gasket.

Note : The word **OUT** must be positioned toward the turbocharger or the raised gasket bead.

Install the turbocharger, washers, and nuts.

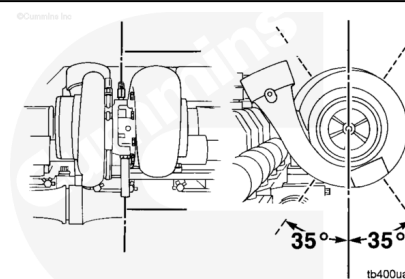
Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]



Position the turbocharger drain tube. The drain tube **must** be within 35 degrees of vertical. Turn the bearing housing to align the tube, if necessary.

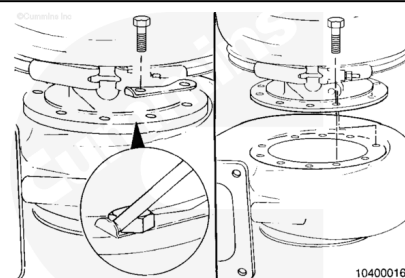
The turbocharger uses capscrews to attach the bearing housing to the turbine housing.



To adjust the bearing housing, bend the lockplates from the capscrew heads.

Remove the capscrews.

Rotate the bearing housing. Align the drain tube and the capscrew holes.

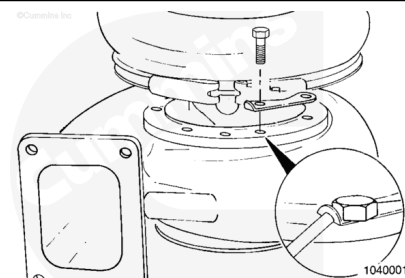


Install the lockplates and capscrews.

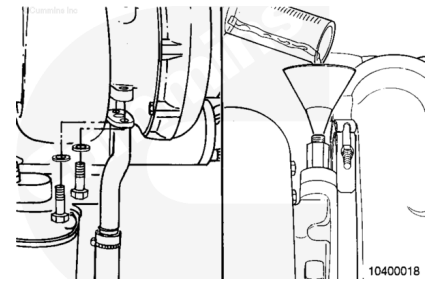
Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]

Bend the lockplate tabs over the capscrews.



These parts are a safety-related item. Be sure the parts are clamped correctly to prevent fretting.



Install the oil drain tube and gasket. Refer to Procedure 010-045 in Section (/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)

Tighten the capscrews.

Torque Value: 45 n·m [33 ft-lb]

Tighten the drain hose swivel to the oil pan adapter.

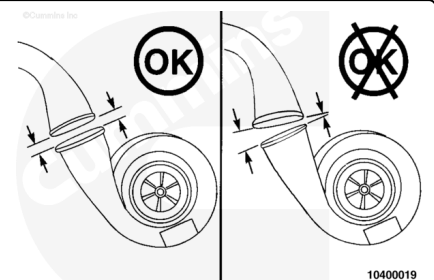
Torque Value: 95 n·m [70 ft-lb]

Pour clean engine oil into the turbocharger oil supply fitting.

Turbocharger Oil Supply: 50 to 60 cc [2.0 to 3.0 oz.]

Check the air crossover-to-turbocharger alignment.

- Install the air crossover on the aftercooler.
- Look at the alignment between the crossover and the turbocharger compressor outlet.
- Remove the crossover and proceed with the assembly if the alignment is correct.
- Proceed to the next step to adjust the compressor housing if the alignment is **not** correct.



To adjust the turbocharger compressor housing alignment:

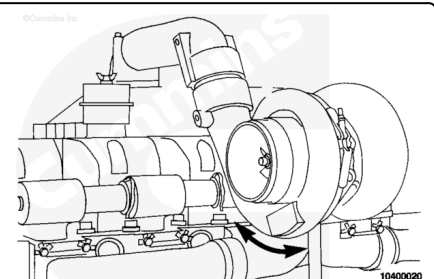
Loosen the v-band clamp.

Turn the compressor housing to the correct alignment.

Tighten the clamp.

Torque Value: 8 n·m [71 in-lb]

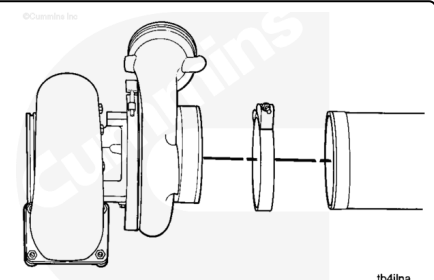
Tap the clamp with a mallet and tighten the clamp again.



Connect the inlet piping to the turbocharger.

Tighten the clamps.

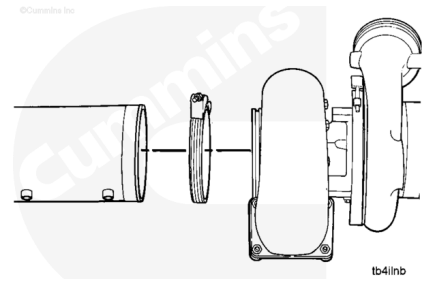
Torque Value: 8 n·m [71 in-lb]



Connect the exhaust piping to the turbocharger.

Tighten the v-band clamp.

Torque Value: 8 n•m [71 in-lb]



Marine Applications

Apply an anti-seize compound to the mounting capscrews.

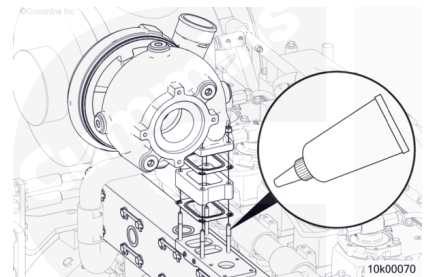
Install the gasket(s).

Note : The word OUT must be positioned toward the turbocharger or the raised gasket bead.

Install the turbocharger, washers, and nuts.

Tighten capscrews in a criss cross pattern.

Torque Value: 40 n•m [30 ft-lb]

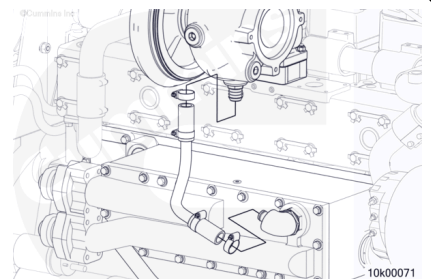


For engines with auxiliary cooler:

Install turbocharger coolant supply line.

Torque Value:

Hose Clamp Torque: 5 n•m [45 in-lb]

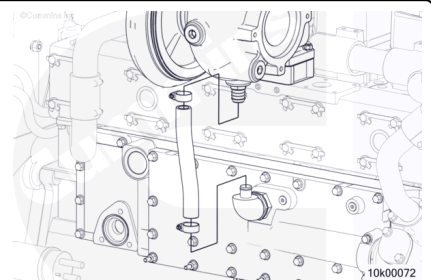


For engines without auxiliary cooler:

Install turbocharger coolant supply line.

Torque Value:

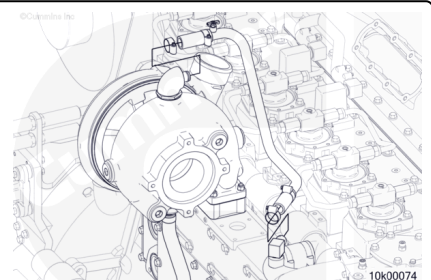
Hose Clamp Torque: 5 n•m [45 in-lb]



Install turbocharger coolant drain line.

Torque Value:

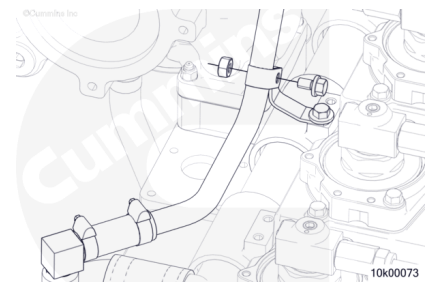
Hose Clamp Torque: 5 n•m [45 in-lb]



Install P-clip

Torque Value:

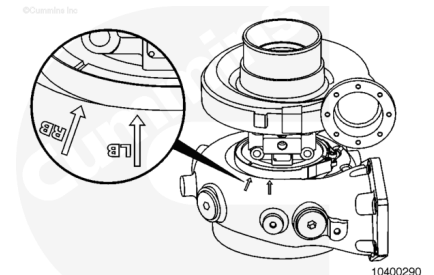
Hose Clamp Torque: 40 n•m [30 ft-lb]



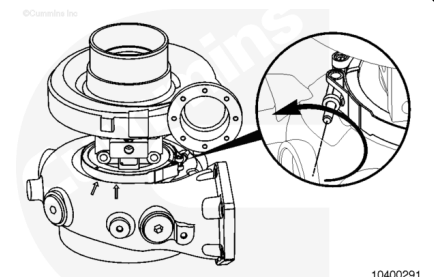
Note : All views showing the clock marks have been simplified for clarity.

Note : The turbocharger for the QSK19 Marine application comes clocked from the factory for the QSK19.

Verify the correct clocking of the turbocharger.

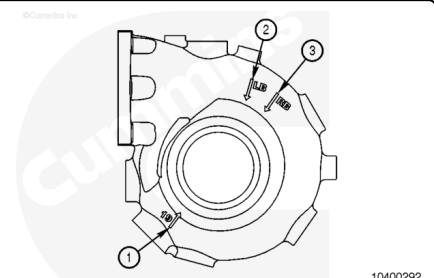


If the turbocharger requires clocking, set the turbocharger on a bench with the turbine housing against the bench and loosen the v-band clamp.

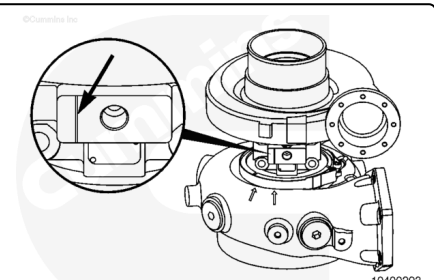


The turbocharger can be used in different applications and has three clock marks on the turbine housing.

1. 19 casting line
2. Left bank
3. Right bank



The bearing housing has a machined surface on the top, creating an alignment mark.



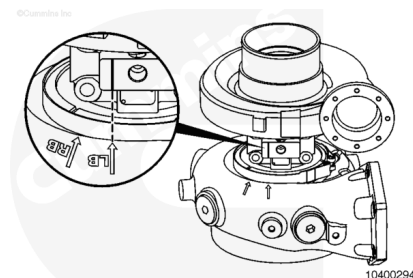
Rotate the bearing housing, aligning the mark with the 19 casting line on the turbine housing.

Apply anti-seize compound to the v-band threads.

Tighten the v-band clamp.

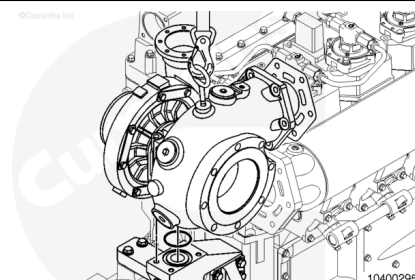
Torque Value:

1. 20 n•m [177 in-lb]
2. Loosen 2 to 3 turns
3. 18 to 20 n•m [159 to 177 in-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



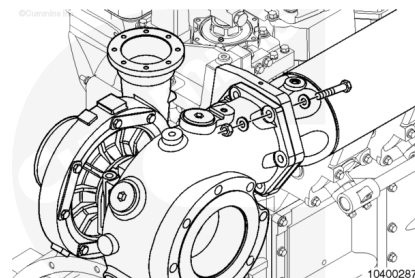
Lubricate and install a new o-ring into the machined groove on the turbocharger center support/drain assembly.

Install the turbocharger and exhaust manifold gasket.

Install the six capscrews, washers, and nuts. Tighten the nuts.

Torque Value:

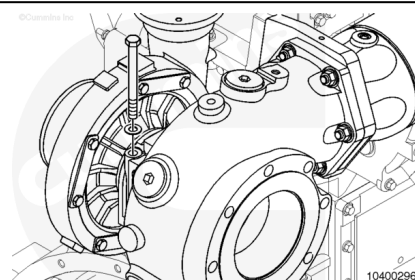
Turbocharger Capscrews 136 n•m [100 ft-lb]



Install the turbocharger to the center section mounting capscrews and washers and tighten.

Torque Value:

Turbocharger to Center Section Mounting Capscrews 115 n•m [85 ft-lb]

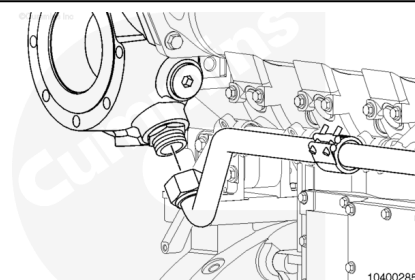


Install the turbocharger turbine housing coolant return line at the turbocharger.

Torque Value: 43 to 47 n•m [32 to 35 ft-lb]

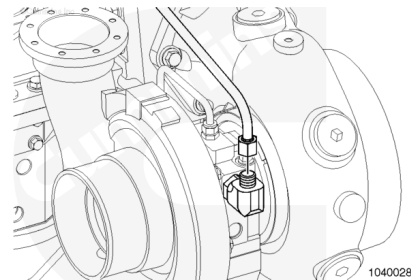
Install a new gasket and capscrews on the coolant bypass tube connection.

Torque Value: 40 n•m [30 ft-lb]



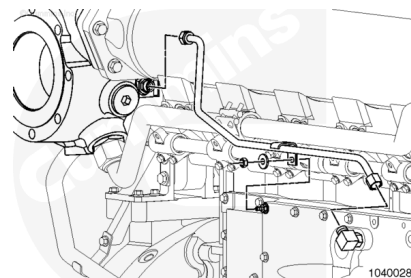
Install the turbocharger bearing housing coolant supply line.

Torque Value: 40 n•m [30 ft-lb]



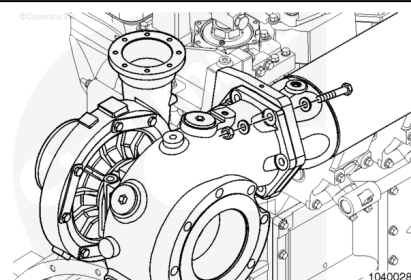
Install the turbocharger bearing housing coolant return line.

Torque Value: 43 to 47 n•m [32 to 35 ft-lb]



Install the three capscrews to the support bracket to compressor housing.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

All Applications Except Marine and Rail

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the coolant supply hoses. Refer to Procedure 010-041 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-041-tr.html)
- Install the oil supply hose. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-046.html)
- Install the oil drain hose. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

- Install the air crossover. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)
- Install the intake and exhaust pipes from the turbocharger. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Marine Applications

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

⚠ CAUTION ⚠

Use caution when filling the oil or replacing filters that the oil is not spilled into the bilge area. Do not pump coolant overboard. If the oil is spilled, it must be disposed of in accordance with local environmental regulations

- Install the turbocharger oil return tube. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-045.html)
- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-046.html)
- Install the exhaust piping. See equipment manufacturer service information.
- Install the air crossover pipe. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html)
- Fill the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Last Modified: 14-Mar-2018
